

DONGJAE LEE

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SUMMARY

I am currently a PhD student at MIT PDOS. My research interests include verification of computer systems, concurrency, and security. I have research experience with the Coq proof assistant, compiler verification, concurrent program verification, and security.

EDUCATION

Ph.D., Computer Science and Engineering	Current
Massachusetts Institute of Technology, Cambridge, MA, USA	
M.S., Computer Science and Engineering	02.2024
Seoul National University, Seoul, South Korea	
• Advised by Chung-Kil Hur • Thesis: Operational Semantics for Expressing and Reasoning about Fairness Properties	
B.S., Physics, Computer Science and Engineering (Double Major)	08.2021
Seoul National University, Seoul, South Korea	

EXPERIENCE

Software Foundations Lab, Seoul National University	03.2024 - 08.2024
Research Assistant: Seoul, South Korea	
Max Planck Institute for Security and Privacy	03.2023 - 08.2023
Research Intern: Bochum, Germany	
• Advised by Cătălin Hrițcu • Worked on secure compilation	
Software Foundations Lab, Seoul National University	09.2020 - 08.2021
Research Intern: Seoul, South Korea	
• Advised by Chung-Kil Hur • Worked on Conditional Contextual Refinement	
ROK Army (Mandatory Military Service)	01.2019 - 08.2020
Sergeant: South Korea	
Integrated Quantum Systems Lab, Seoul National University	04.2017 - 08.2017
Research Intern: Seoul, South Korea	
• Advised by Dohun Kim • Worked on NV center qubits, programming and developing devices for experiments	

PUBLICATIONS

*equal contribution

Lilo: A Higher-Order, Relational Concurrent Separation Logic for Liveness
Dongjea Lee, Janggun Lee, Taeyoung Yoon, Minki Cho, Jeehoon Kang, Chung-Kil Hur.
International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA 2025)

Refinement Composition Logic
Youngju Song, Dongjea Lee.
International Conference on Functional Programming (ICFP 2024)

SECOMP: Formally Secure Compilation of Compartmentalized C Programs
Jérémy Thibault, Roberto Blanco, Dongjae Lee, Sven Argo, Arthur Azevedo de Amorim, Aïna Linn Georges, Cătălin Hrițcu, Andrew Tolmach.
Conference on Computer and Communications Security (CCS 2024)

Stuttering for Free

Minki Cho*, Youngju Song*, **Dongjae Lee**, Lennard Gäher, Derek Dreyer.

International Conference on Object-Oriented Programming, Systems, Languages, and Applications (OOPSLA 2023)

Fair Operational Semantics

Dongjae Lee*, Minki Cho*, Jinwoo Kim, Soonwon Moon, Youngju Song, Chung-Kil Hur.

Conference on Programming Language Design and Implementation (PLDI 2023)

Conditional Contextual Refinement

Youngju Song, Minki Cho, **Dongjae Lee**, Chung-Kil Hur, Michael Sammler, Derek Dreyer.

Symposium on Principles of Programming Languages (POPL 2023)

Sequential Reasoning for Optimizing Compilers under Weak Memory Concurrency

Minki Cho*, Sung-Hwan Lee*, **Dongjae Lee**, Chung-Kil Hur, Ori Lahav.

Conference on Programming Language Design and Implementation (PLDI 2022)

HONORS AND AWARDS

Master's Thesis Award	02.2024
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Department of Computer Science and Engineering, Seoul National University: Seoul, South Korea

TALKS

Lilo: A Higher-Order, Relational Concurrent Separation Logic for Liveness	10.2025
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OOPSLA 2025: Singapore

Fair Operational Semantics	06.2023
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PLDI 2023: Orlando, Florida, United States

Overview of Fair Operational Semantics (as a part of introducing Software Foundations Lab)	02.2023
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SIGPL Winter School 2023 (The Korean Institute of Information Scientists and Engineers): Seoul, South Korea

TEACHING

(TA) Topics in Programming Languages (Logic in computer science)	09.2023 - 12.2023
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by Makoto Tatsuta: Seoul National University, Graduate level course

(TA) Principles of Programming	09.2022 - 12.2022
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by Chung-Kil Hur: Seoul National University, Undergraduate level course

(TA) Principles and Practices of Software Development	03.2022 - 06.2022
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by Chung-Kil Hur: Seoul National University, Undergraduate level course

ACTIVITIES

A Rocq tutorial for refinement-based verification: <https://github.com/dongjaelee1/refinement-tutorial>